Customer Segmentation and Purchase Prediction using Machine Learning

This project focuses on analyzing customer purchasing behavior using machine learning techniques. The primary goals were to perform customer segmentation using clustering algorithms and predict future customer spending using regression models. The project was implemented using Python, Pandas, Scikit-learn, Matplotlib, and Seaborn.

1. Data Cleaning

The dataset contained customer transaction information including CustomerID, Invoice details, Quantity purchased, Product information, UnitPrice, and InvoiceDate. Several preprocessing steps were performed to ensure high data quality before model building.

- Removed rows containing missing CustomerID values.
- Removed duplicate transactions from the dataset.
- Filtered negative quantities and invalid prices.
- Converted InvoiceDate into datetime format.
- Created TotalAmount feature using Quantity × UnitPrice.
- Handled NaN and infinite values generated during feature engineering.

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand customer purchasing patterns, distribution of spending, transaction frequency, and customer engagement. Several visualizations were created using Matplotlib and Seaborn.

- Distribution plots for customer spending behavior.
- Frequency analysis of customer purchases.
- Scatter plots between Frequency and Monetary value.
- Cluster visualization using color-coded customer groups.
- Heatmaps showing average cluster characteristics.

3. Feature Engineering

Feature engineering was an important step in improving the predictive performance of machine learning models. Several business-oriented customer features were created.

Feature	Description
Recency	Days since last purchase
Frequency	Number of customer purchases
Monetary	Total customer spending
AvgOrderValue	Average spending per transaction
SegmentEncoded	Encoded customer segment labels

Cluster	K-Means cluster assigned to each customer

4. Customer Segmentation using K-Means

Customer segmentation was performed using the K-Means clustering algorithm. Customers were grouped based on Recency, Frequency, and Monetary values. The Elbow Method was used to determine the optimal number of clusters.

- Cluster 0 – High-value customers with strong spending behavior.
- Cluster 1 – Regular customers with moderate activity.
- Cluster 2 – Occasional customers with lower engagement.

5. Predictive Modeling

Three regression models were implemented to predict future customer spending behavior. The target variable was transformed using logarithmic scaling to reduce skewness.

Model	Purpose
Linear Regression	Baseline linear prediction model
Decision Tree Regressor	Captures non-linear customer behavior
Random Forest Regressor	Ensemble model with improved prediction accuracy

6. Model Evaluation

The models were evaluated using RMSE (Root Mean Squared Error) and R^2 Score. The Random Forest model achieved the best performance among all models.

Model	RMSE	R^2 Score
Linear Regression	0.767	0.512
Decision Tree	0.682	0.614
Random Forest	0.654	0.645

7. Business Insights

The project generated several actionable business insights that can help organizations improve customer retention and increase revenue.

- High-frequency customers contribute significantly to total revenue.
- Customer segmentation helps businesses personalize marketing campaigns.
- Random Forest provided the most accurate prediction performance.
- Recent and frequent buyers are more likely to generate future purchases.
- Customer analytics can improve retention strategies and loyalty programs.

8. Conclusion

This project successfully demonstrated an end-to-end machine learning workflow including data cleaning, exploratory data analysis, feature engineering, customer segmentation, predictive modeling, and business insight generation. The final Random Forest model showed strong predictive capability and can be effectively used for customer analytics and personalized marketing strategies.